Task: Evaluating Autonomous Pick-and-Place Using a Simulated SO-ARM100[1] Robot in Webots

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A. HIGHLIGHTS

Question: Can a robot-control policy trained from a small set of demonstrations in one simulated environment, using only simple numeric inputs (no camera), still work when the environment changes, without retraining?

Role & Contributions: I built the data collection pipeline (a teleoperation script and a module that randomizes object positions), designed the 14-value state representation, collected 300 expert demonstrations, and trained a diffusion policy on them using an existing implementation. I then built a runner to deploy the policy in four new simulated environments and evaluated it over 250 trials (50 per environment, including the training environment). I did the data analysis: success rates, confidence intervals, weighted mean completion times, and used Google Sheets for inputting data and generating charts.

Main Findings: The policy completed the task on its own in 0 of 250 trials because it did not learn to grasp. With manual grasp assistance, which let me test the rest of the task, success fell from 90% in the training environment to 44% when changes were combined, and lower success went with longer completion times. With 50 trials per environment, the three single-change environments (62% to 70%) could not be ranked. The 14-value state gives joint positions and box and goal poses but no direct gripper-to-object offset or contact feedback, which is a plausible explanation for the grasp failure.

B. CONTEXT

1. OVERVIEW

This document presents the experimental results and quantitative evaluation from my undergraduate thesis which utilized Low-Dimensional Diffusion Policy[2] to teach SO-ARM100 pick-and-place behaviors using Webots[3] demonstrations.

The said study evaluated an existing low-dimensional Diffusion Policy implementation across five simulated environments to examine its pick-and-place performance under different environmental conditions. Expert pick-and-place demonstrations were generated first, which served as data to train the policy on. The evaluation covered 250 trials, comprising 50 trials per environment.

The analysis focused on task success rates, completion times, failure patterns, and performance changes under different environmental variations.

2. METHODOLOGY

Input to train the policy:
Via teleoperation, I generated 300 experts demos and extracted the ff:

- State (14 values): 6 joint positions (5 arm joints and the gripper), the box pose (x, y, z, yaw), and the goal pose (x, y, z, yaw). This is what served as the input.

[1] SO-ARM100 is an open-source 6 DOF robotic arm designed for AI and robotics research developed by The Robot Studio in collaboration with Hugging Face as part of the LeRobot project
[2] Diffusion Policy (DP) is a robot learning method that generates actions by iteratively denoising random noise, enabling robots to learn complex behaviors from demonstrations.
[3] Webots is an open source robotic simulation platform widely used in navigation, manipulation, and swarm robotics.

Environments:

The trained policy was evaluated across five Webots environments: the data collection world plus the four novel configurations with 50 trials per environment (250 total). The five environments are the ff:

1. Collection environment: The environment used for demonstration collection
2. Larger workspace: An environment with expanded workspace boundaries
3. Novel object orientations: An environment testing object orientations beyond the original training distribution
4. Novel robot initial states: An environment with variations in the robot's starting configuration
5. Combined variations: An environment incorporating multiple changes simultaneously

Two performance measures were distinguished. I added an assisted measure after finding that the policy had not learned to grasp

a. Fully autonomous success: The robot independently completed the entire pick-and-place task, including grasping the object and placing it within the designated goal zone.

b. Assisted success: The policy autonomously positioned the gripper above the object, after which an operator manually assisted with grasping. The policy then autonomously transported and placed the object within the goal zone.

C. DATA: TABLES, PROCESSING, AND CALCULATIONS

3. RESULTS

3.1 Fully Autonomous Performance

The policy achieved a 0% success rate across all 250 trials. The observed failures occurred during grasp initiation. Although the policy positioned the end-effector above the object and opened the gripper, it did not reliably establish a successful grasp. These results indicate that the policy, under the evaluated setup, was unable to complete the full pick-and-place task without intervention.

One possible explanation is that the state representation did not explicitly encode relative gripper-to-object geometry or provide tactile and contact-force feedback. However, the experiments did not establish this as the cause of failure. Further controlled testing would be needed to determine which factors contributed most to the grasping failures.

3.2 Performance With Manual Grasp Assistance[4]

When manual grasp assistance was provided, in terms of success rate and completion time, the policy achieved the following in each inference environment:

[4] This is when the autonomous policy activity is intentionally halted and I take over the control via keyboard (using the intervention teleoperation script) to control the gripper’s movement, then after successful grip, the autonomous policy is resumed.

Policy Performance in Each Inference Environment															
Data Collection World			Larger Workspace			Novel Box Orientations			Novel Robot Arm Initial State			All Variations Combined			
Trial	Successful?	Completion Time (s)	Trial	Successful?	Completion Time (s)	Trial	Successful?	Completion Time (s)	Trial	Successful?	Completion Time (s)	Trial	Successful?	Completion Time (s)	
1	1	70.76	1	0	-	1	0	-	1	1	97.26	1	0	-	
2	1	73.65	2	1	98.42	2	0	-	2	0	-	2	0	-	
3	1	66.95	3	1	112.76	3	1	79.42	3	1	90.73	3	0	-	
4	1	70.59	4	0	-	4	1	91.68	4	0	-	4	1	114.22	
5	1	63.64	5	1	119.35	5	1	84.75	5	1	102.8	5	1	106.62	
6	1	68.61	6	1	101.63	6	0	-	6	0	-	6	1	116.55	
7	0	-	7	1	104.18	7	1	96.13	7	1	92.5	7	0	-	
8	1	63.91	8	0	-	8	1	87.24	8	1	85.26	8	1	103.56	
9	1	68.86	9	1	108.91	9	0	-	9	1	100.71	9	0	-	
10	0	-	10	0	-	10	0	-	10	1	94.58	10	0	-	
11	1	68.02	11	1	95.84	11	0	-	11	0	-	11	1	117.43	
12	1	71.35	12	1	115.27	12	1	82.91	12	1	99.07	12	0	-	
13	1	67.03	13	1	106.48	13	1	89.56	13	1	87.02	13	0	-	
14	1	64.99	14	0	-	14	1	76.83	14	0	-	14	1	100.23	
15	1	65.1	15	1	110.12	15	1	93.17	15	0	-	15	1	115.66	
16	1	62.03	16	1	103.75	16	0	-	16	1	101.59	16	0	-	
17	1	74.69	17	0		17	0	-	17	1	82.9	17	1	105.39	
18	1	61.13	18	1	121.06	18	1	85.49	18	1	99.52	18	0	-	
19	1	68.59	19	0	-	19	1	90.42	19	1	96.73	19	0	-	
20	1	71.23	20	0	-	20	0	-	20	1	85.71	20	0	-	
21	1	67.8	21	0	-	21	1	83.67	21	0	-	21	1	117.89	
22	1	63.09	22	0	-	22	1	98.21	22	0	-	22	1	102.47	
23	1	74.05	23	1	99.87	23	0	-	23	0	-	23	0	-	
24	1	70.09	24	1	107.24	24	1	86.54	24	1	97.61	24	1	113.67	
25	1	73.62	25	1	113.68	25	1	80.76	25	1	88.56	25	1	111.38	
26	1	62.86	26	0	-	26	0	-	26	1	98.58	26	0	-	
27	1	69.01	27	1	102.39	27	1	88.93	27	1	103.86	27	0	-	
28	1	71.6	28	0	-	28	1	92.35	28	1	91.81	28	0	-	
29	0	-	29	0	-	29	1	84.12	29	0	-	29	0	-	
30	1	79.2	30	1	109.56	30	1	77.94	30	1	105.17	30	1	118.98	
31	1	68.76	31	1	96.71	31	1	89.18	31	1	93.27	31	1	109.05	
32	1	69.49	32	1	117.42	32	0	-	32	1	89.29	32	1	122.04	
33	1	74.64	33	0	-	33	1	95.62	33	0	-	33	0	-	
34	1	63.08	34	1	105.93	34	1	81.37	34	1	95.44	34	0	-	
35	1	67.03	35	1	111.84	35	1	87.61	35	1	92.3	35	1	108.17	
36	1	60.93	36	1	100.56	36	1	90.87	36	1	100.15	36	1	125.37	
37	0	-	37	0	-	37	1	85.06	37	1	94.15	37	0	-	
38	1	67.05	38	1	114.91	38	1	99.14	38	1	102.32	38	1	109.94	
39	1	65.55	39	0	-	39	1	83.28	39	0	-	39	0	-	
40	1	66.3	40	1	103.12	40	0	-	40	1	90.17	40	0	-	
41	1	71.72	41	1	106.82	41	1	88.46	41	1	98.38	41	0	-	
42	1	71.31	42	1	118.24	42	1	79.85	42	0	-	42	0	-	
43	0	-	43	1	97.65	43	0	-	43	1	105.62	43	1	120.21	
44	1	70.28	44	0	-	44	1	91.24	44	1	96.3	44	1	107.71	
45	1	67.26	45	1	110.73	45	0	-	45	1	88.08	45	0	-	
46	1	66.02	46	1	104.86	46	0	-	46	1	91.36	46	0	-	
47	1	63.13	47	0	-	47	1	86.92	47	0	-	47	1	123.13	
48	1	69.76	48	0	-	48	1	94.38	48	1	107.98	48	0	-	
49	1	70.6	49	1	111.86	49	1	88.84	49	0	-	49	1	111.93	
50	1	68.14	50	1	108.37	50	0	-	50	1	100.15	50	0	-	
	45	68.3		31	107.73		33	87.63		35	95.63		22	112.8	
	SUM	MEAN		SUM	MEAN		SUM	MEAN		SUM	MEAN		SUM	MEAN	

View sheet: Data and charts

Table 3.1 Summary of Results

Inference World	Trials	Unsuccessful	Successful	Success Rate	Confidence Interval [5]	Mean Completion Time (s)
1. Collection World	50	5	45	90%	79% to 96%	68.3
2. Larger Workspace	50	19	31	62%	48% to 74%	107.73
3. Novel box orientations	50	17	33	66%	52% to 78%	87.63
4. Novel robot initial state	50	15	35	70%	56% to 81%	95.63
5. Combined Variations	50	28	22	44%	31% to 58%	112.8
Overall	**250**	**84**	**166**	**(Mean) 66.4%**	**60% to 72%**	**(Weighted Mean) 91.17**

Table 3.2 Summary of Success Rates

Inference World	Trials	Success Rate	
		Policy	Manually-Assisted
Collection World	50	0%	90%
Larger Workspace	50	0%	62%
Novel box orientations	50	0%	66%
Novel robot initial state	50	0%	70%
Combined Variations	50	0%	44%
Overall	**250**	**0%**	**(Mean) 66.4%**

The collection environment produced the highest assisted success rate at 90%. Among the individual novel conditions, the robot-initial-state variation achieved the highest rate at 70%, followed by novel object orientations at 66% and the larger workspace at 62%. The combined-variation environment had the lowest rate at 44%.

These results suggest that the policy performed better under individual environmental changes than when multiple changes were introduced together. However, the results alone do not establish which variation caused the observed differences.

Table 3.3 Mean Completion Time Summary

Inference World	Trials	Mean Completion Time (s)
Collection World	50	68.3
Larger Workspace	50	107.73
Novel box orientations	50	87.63
Novel arm initial state	50	95.63
Combined Variations	50	112.8
Overall	**250**	**(Weighted Mean[6]) 91.17**

NOTE: The environment means are already rounded, so we treat 91.17 seconds, the weighted mean, as an approximation.

From the tables it can be inferred that individual changes cost less than combined ones. Each single change lowered success from 90% to between 62% and 70%, while combining them lowered it to 44%. The middle three environments are statistically indistinguishable. Their confidence intervals overlap heavily, so I cannot say which single change hurts most. Only the collection environment (high) and the combined environment (low) clearly differ.

[5] The range of true success rates that fit what is observed. A 95% CI is a standard way to show how much a rate measured on a limited number of trials could vary.

[6] Weighted mean is an average that accounts for differences in the number of observations in each group.

Success and speed move together. Across the five environments, higher success rates went with shorter completion times (correlation r ≈ -0.93). This rests on only five points, and the times come only from successful trials, so I treat it as a pattern consistent with harder environments being both less reliable and slower, not as a proven relationship.

3.3 Data Processing and Calculations

The collected trial results were summarized using success counts, success rates, and mean completion times for each test environment.

1. Success Rate

   The success rate was calculated by dividing the number of successful trials by the total number of trials and multiplying the result by 100.

$$Success\ rate_{i} = \frac{Number\ of\ successful\ trials\ in\ environment\ i}{50}\ x\ 100\%$$

2. Completion Time

$$T_{completion} = Time\ from\ episode\ start\ to\ successful\ placement$$

3. Mean Completion Time

   The mean completion time represents the arithmetic average of the recorded completion times for the trials included in the calculation. Completion times were summarized by environment to compare task duration across the different test conditions.

$$\bar{x} = \frac{Sum\ of\ all\ Completion\ Time\ in\ the\ environment}{Total\ number\ of\ successful\ trials}$$

4. Weighted Mean for Overall Mean Completion Time

   This is used to calculate the overall completion time when each environment has a different number of successful trials which is the case here.

$$\bar{x}_{overall} = \frac{\sum_{j=1}^{k} n_j \bar{x}_j}{\sum_{j=1}^{k} n_j}$$

Where:

n_j = number of successful trials in environment *j*

$\bar{x}_j$ = mean completion time for that environment

k = number of environments

5. Confidence Interval (CI) for Success Rate

   The 95% confidence intervals for success rates were calculated using the Wilson score interval, which estimates uncertainty in the observed success proportion.

   Let $p = x/n$, where x is the number of successful trials and n is the total number of trials. The Wilson score interval is calculated as:

$$\frac{p + \frac{z^2}{2n} \pm z\sqrt{\frac{p(1-p)}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

Where:

p = observed success proportion (successes / trials)

x = number of successful trials

n = total number of trials

z = critical value; 1.96 for a 95% confidence interval

NOTE: The environment means are already rounded, so we treat 91.17 seconds, the weighted mean, as an approximation.

D. VISUALIZATION

4. DATA VISUALIZATION

Figure 4.1 Results Summary

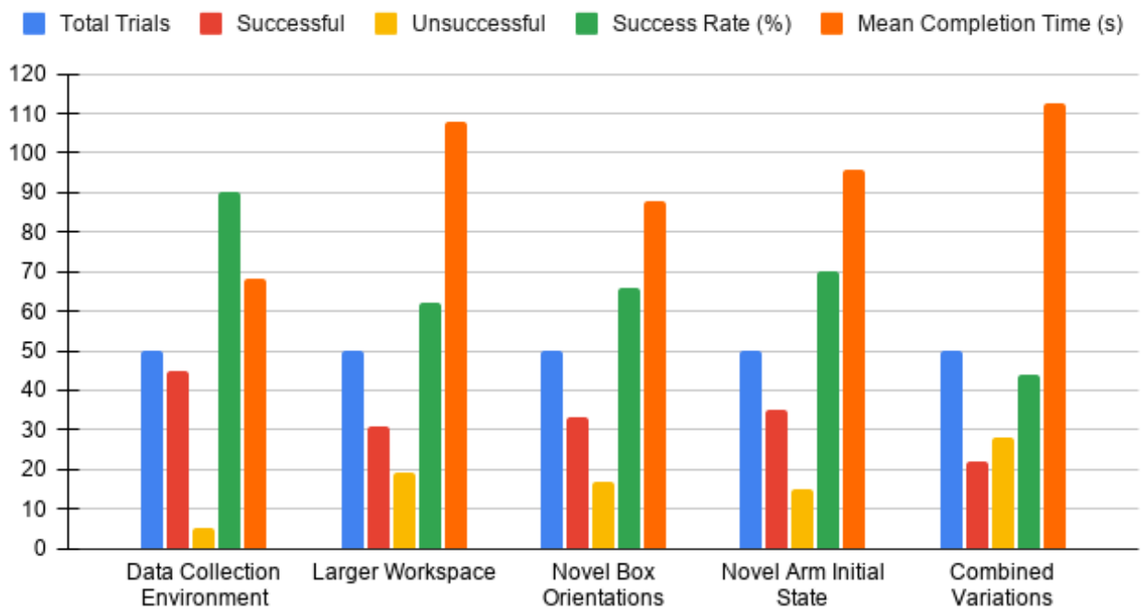
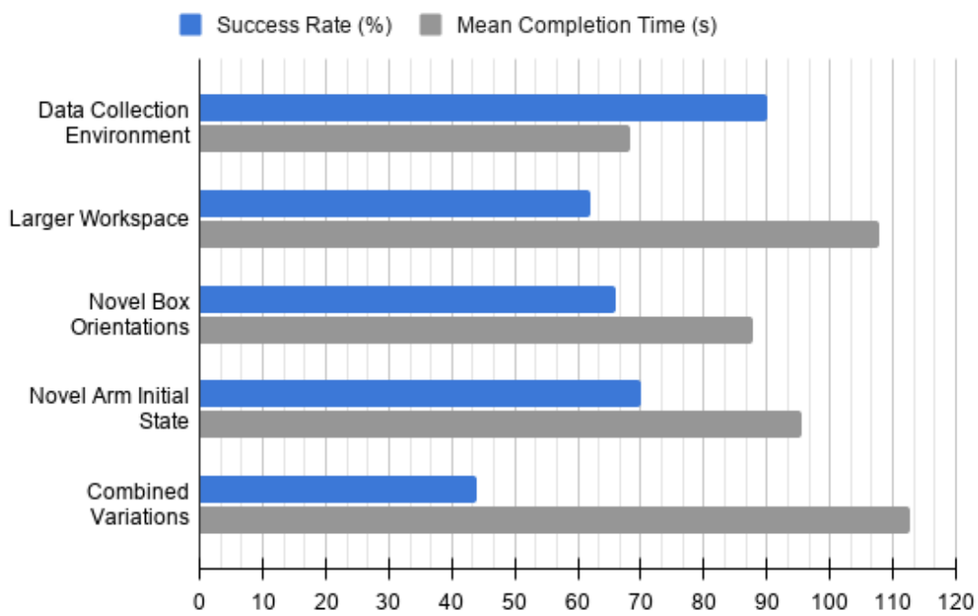
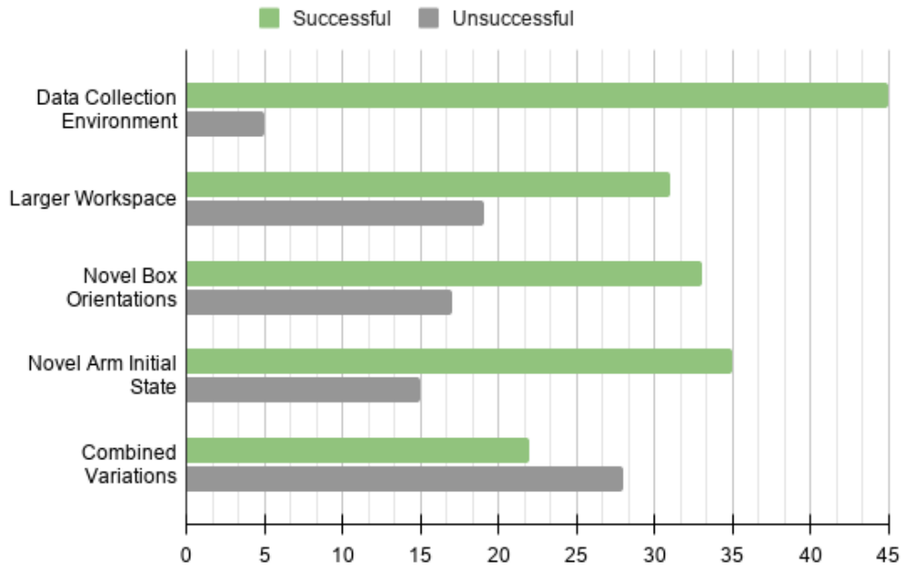


Figure 4.2 Success Rate and Mean Completion Time Comparison



Across the five environments, higher assisted success rates generally corresponded to shorter mean completion times, while lower success rates tended to coincide with longer completion times. The collection environment recorded the highest success rate (90%) and shortest mean completion time (68.30 seconds), whereas the combined-variations environment recorded the lowest success rate (44%) and longest mean completion time (112.80 seconds). This pattern suggests that environments with more challenging conditions were associated with both reduced task success and longer execution times.

Figure 4.3 Successful Trials VS Unsuccessful Trials in Each Environment



As the test conditions became more varied, the results showed a general shift toward more unsuccessful trials. The collection environment had the fewest failures (5) and most successes (45), while the combined-variations environment had the most failures (28) and fewest successes (22). This pattern is consistent with reduced task reliability under combined variations.

E. KEY FINDINGS

This evaluation has produced 7 main findings:

1. The policy did not learn to grasp. It had 0 successes in 250 fully autonomous trials (95% upper bound about 1.5%), and every failure occurred at grasp initiation.

2. Performance varied across environmental conditions. Assisted success rates ranged from 44% to 90%, with the lowest rate occurring when multiple environmental variations were combined.

3. Environmental changes affected observed performance. The results show differences in success rates and completion times.

4. Lower success went with longer completion times ($r \approx -0.93$ across the five environments). Five points and successful-trial times only, so this is a pattern, not a proven relationship.

5. The three single-change environments cannot be ranked. At 62%, 66%, and 70%, their confidence intervals overlap, so 50 trials per environment cannot show which change hurts most.

6. The 14-dimensional state representation may not have been sufficient to encode relative gripper-to-object geometry for the policy to learn grasping.

7. Despite fewer expert demonstrations (300 episodes only), Diffusion Policy was able to learn navigation: gripper to object to the goal zone.

F. LIMITATIONS

1. Why the grasp failed is untested. My working explanation is that the 14-dimensional state gives the box pose and joint positions but no direct gripper-to-object offset or contact feedback, so the policy would have to infer grasp timing indirectly. I did not test this (for example, by adding the relative offset to the state), so it remains a hypothesis, and other factors, such as having only 300 demonstrations, were not ruled out.

2. Assisted results include human input, so they describe the rest of the task, not a working system.

3. Completion times cover successful trials only and include my manual grasp, so environments with fewer successes are averaged over fewer trials.

4. Five data points limit any statement about the relationship between success and time.

5. Simulation only. Results may not carry over to a physical robot.

G. CONCLUSION

A policy trained on 300 demonstrations in one simulated environment learned to move the gripper to the object and the goal zone, but it did not learn to grasp: it succeeded in 0 of 250 autonomous trials. With manual grasp assistance, success depended on the environment, falling from 90% where the policy was trained to 44% when several changes were combined. With 50 trials per environment, only the larger differences were clear; the single-change environments could not be ranked.

The most useful next step is to test the grasp explanation directly by adding the relative gripper-to-object offset to the state, retraining, and rerunning the autonomous trials. Further steps are to test one variation at a time with more trials and to add demonstrations from the varied environments.